

JOHN JAY COLLEGE OF CRIMINAL JUSTICE
The City University of New York
524 West 59th Street, New York, NY, 10019

Syllabus for FOS 709
Applied Statistics and Data Analytics for the Laboratory Forensic Sciences

Professor's name: Nicholas Petraco

Semester: Spring 2026

Course Code: FOS709

Lecture location/time: 5.67, M 3:45pm-5:45pm New Building

Laboratory location/time: 5.67, W 4:00pm-7:00pm New Building

Office Contact hours: 5.66.10 New Building, W 3:00pm-4:00pm and Open Door Policy

E-mail address: npetraco@gmail.com, npetraco@jjay.cuny.edu, nipetraco@jjay.cuny.edu

Course website: <https://npetraco.github.io/709/>

Course Prerequisite: None

Americans with Disabilities Act (ADA) Policies – CUNY Accommodations Policy:

Students who need an accommodation due to a disability are encouraged to contact the Office of Accessibility Services (OAS) within the first week of class or as soon as possible thereafter. Accommodations can only be approved by the OAS and will be implemented for the student. OAS is located at L66 in the new building, Phone: (212-237-8031),

Email: accessibilityservices@jjay.cuny.edu. See CUNY ADA policies for students: [Effecting Reasonable Accommodations and Academic Adjustments Procedures Relating to Accommodations and Accessibility for Students](#)

(<https://www.cuny.edu/about/administration/offices/legal-affairs/policies-procedures/reasonable-accommodations-and-academic-adjustments/>)

Course description:

Tailored for Forensic Science Masters degree students, this practical introductory course in statistics and data analytics combines theory with hands-on training. Designed to address the practicalities of analyzing and modeling forensic laboratory data for research and practice, the course emphasizes rigorous model selection, testing, and criticism. Lectures and software training/laboratory components are integrated, utilizing datasets from trace evidence, toxicology, and molecular biology/DNA analysis.

Course Focus:

It is not adequate to simply learn the “theory” behind the statistical methodology encountered in this course. It must be applied on real data of practical interest to the sciences. As such this course will build expertise in the general scientific/statistical computing environment R (<http://www.r-project.org/>). The course assumes minimal knowledge of computers and statistical procedures. It is designed to build the student’s skill set and confidence in both areas. It will address statistical methods that are applicable to data students will encounter in their scientific careers, as well as issues and pitfalls to be aware of when applying these methods. Students must be in possession of a laptop running Windows, Mac OS or Linux for this class. If a student is not in possession of a laptop, one can be borrowed from the school. A tablet will not suffice for this course. Topics covered will include basic data descriptive tools, graphing, distributions, interval estimates, hypothesis testing, topics in experimental design, regression models, model criticism, introductory Bayesian analysis, and statistical standards resources.

Course lecture materials will be posted on the following website:

<https://npetraco.github.io/709/>

Learning outcomes:

By the end of the course students will be able to:

- Formulate scientific questions to be answered using data from forensic laboratory experiments and appropriate statistical methodologies.
- Demonstrate an understanding of what conditions/assumptions are typically required for the use of specific statistical models; and be able to assess whether those conditions are reasonably met under a set of given experimental conditions.
- Draw appropriate inferences from outputs of particular statistical models.
- Demonstrate skill in using the R software system to handle/process datasets, display data graphically and perform the statistical analysis pipelines covered in the course.
- Demonstrate the importance of accuracy and objectivity in ethically collecting scientific data and drawing inferences from it.

Required Text and Electronic Resources:

Text:

Foundations of Statistics for Data Scientists: With R and Python: (FS)

A. Agresti and M. Kateri

Website: <https://stat4ds.rwth-aachen.de/>

ISBN-10: 0367748452

WebAssign:

<https://www.webassign.net/>

Class Keys:	Lab Section 01
	jjay.cuny 6889 8998

Computer:

A laptop running Windows, Mac OS or Linux operating systems is required for this course. If a student is not in possession of a laptop, one can be borrowed from the school. A tablet or Microsoft surface will not suffice for this course.

Supplementary texts and resources:

The R Book: (RB)

MJ Crawley

ISBN-10: 8126569719

A Course in Statistics with R: (CSR)

PN Tattar, S Ramaiah, BG Manjunath

ISBN-10: 1119152720

OpenIntro Statistics:

Website: <https://www.openintro.org/book/os/>

Statistics (LibreTexts):

<https://stats.libretexts.org/Bookshelves>

Bureau International des Poids et Mesures - Guide to Measurement Uncertainty: (GUM)

<https://www.bipm.org/en/committees/jc/jcgm/publications>

NIST/SEMATECH e-Handbook of Statistical Methods: (NS)

<https://www.itl.nist.gov/div898/handbook/>

<https://doi.org/10.18434/M32189>

NIST Forensic Statistics: (NFS)

<https://www.nist.gov/spo/forensic-science-program/evidential-statistics>

EPA QA/G-9S: Data Quality Assessment: Statistical Methods for Practitioners: (EPA)

<https://www.epa.gov/sites/default/files/2015-08/documents/g9s-final.pdf>

FDA Investigating Out of Specification Test Results: (FDA)

<https://www.fda.gov/media/158416/download>

FDA Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests

<https://www.fda.gov/media/71147/download>

FDA E8 General Considerations for Clinical Studies: Guidance for Industry

<https://www.fda.gov/media/157560/download>

Requirements/course policies:

Course announcements and important reminders will be discussed in class and emailed to you. **As such you must give me email addresses that you check on a regular basis, including your John Jay email.** Home work, labs and exams will be administered through WebAssign. See below for details.

Students must check the course website and the e-mail account(s) they gave for this course regularly.

Students are responsible for all course information, assignments, announcements, and communication that occurs in class, through the course website and your email accounts.

Students must be in possession of a laptop running Windows, Mac OS or Linux for this class.

If a student is not in possession of a laptop, one can be borrowed from the school. A tablet or Microsoft surface will not suffice for this course. Students are responsible for being in possession of a laptop installed with the course software (R <https://www.r-project.org/> and RStudio <https://posit.co/download/rstudio-desktop/>) before *each and every* lecture and laboratory session.

Attendance in lecture and laboratory is mandatory. More than five unexcused absences from any of these components will result in an automatic failing grade. Unexcused lateness or early departure will count as ½ an absence, up to 30 minutes. After 30 minutes students will be marked absent.

Unethical/unprofessional conduct which includes cheating will result in a failing grade and referral for additional action. These include copying others' work and sharing work when explicitly forbidden. Exams must be taken in person, in class. **No make up exams will be given without prior instructor approval.** Failure to take a scheduled examination in person, in class without valid and independently supported official documentation from a medical provider at least 24 hours in advance (unless the emergency is induced by force majeure, subsequent to the 24 hour cutoff, where in valid and independently supported official documentation from a medical provider is still required) will result in a zero grade for that examination.

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See CUNY ADA policies for students: [Effecting Reasonable Accommodations and Academic Adjustments Procedures Relating to Accommodations and Accessibility for Students](https://www.cuny.edu/about/administration/offices/legal-affairs/policies-procedures/reasonable-accommodations-and-academic-adjustments/) (<https://www.cuny.edu/about/administration/offices/legal-affairs/policies-procedures/reasonable-accommodations-and-academic-adjustments/>)

Statement of the College Policy on Plagiarism:

“Plagiarism is the presentation of someone else’s ideas, words, or artistic, scientific, or technical work as one’s own creation. Using the ideas or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations require citations to the original source.

Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism. It is the student’s responsibility to recognize the difference between statements that are common knowledge (which do not require documentation) and restatements of the ideas of others. Paraphrase, summary, and direct quotation are acceptable forms of restatement, as long as the source is cited.

Students who are unsure how and when to provide documentation are advised to consult with their instructors. The Library has free guides designed to help students with problems of documentation.”

Policy and Source Material:

<http://johnjay.jjay.cuny.edu/files/cunypolicies/JohnJayCollegePolicyofAcademicIntegrity.pdf>

Grading:

Exams

There will be two regular exams and a final (exam III). Regular exams will take place during the lab period to minimize time pressure. The final (exam III) will take place during the scheduled time, finals week. Exam II and the final (exam III) are semi-cumulative in that all exams build on the concepts of the previous exams. Concepts and methods from the earlier parts of the semester will appear on exams during later parts of the semester. That said, reviews before each exam will be thorough, and each exam will emphasize material that has not yet been tested. All exams must be taken in person, in class. Failure to take an exam in person in class without prior instructor approval and/or arrangements will result in an automatic grade of zero for that exam. Each exam (*i.e.* exam I, II and III) will be worth 20% of the total class grade. See course policy above for missed exams.

Workshops/Laboratories

There will be a workshop and/or computational laboratory tutorial exercises for each lab period to accompany lecture. The workshops and laboratory tutorials train students on standard statistical analysis tasks for the lab sciences and research. They are designed to make the concepts in lecture concrete and use lab relevant datasets as a medium. The R language and RStudio software will be used throughout. Participation/attendance in the laboratory tutorial workshops and exercises will be worth 10% of the total class grade. No attendance will be marked as a zero grade.

Home Work Sets

Approximately each week there will be a short home work set consisting of exercises which reenforce and illustrate the material being discussed in the lecture/workshops and lab tutorials. They are due one week after they have been assigned. Collectively the homework exercises will

be worth 30% of the total class grade. Home works up to one week late can be turned in for a 25% penalty. After one week of unexcused lateness, HW sets will receive a zero grade.

In summary, the grade for this course will be based on two exams (40%), a final (20%), weekly homework sets (30%) and workshop/laboratory attendance/participation (10%).

Course Schedule:

Week	Lecture Topics	Readings	Tutorial Lab Topics	Notes/Exams	HW Set Assignment	HW Set Due
Jan.26, Jan.28	Intro, Definitions Resources.	FS Ch. 1	Workshop: Intro to software for the course			
Feb.2, Feb 4	Essential Graphical Tools	FS Ch. 1	Workshop: Intro to R			
Feb.9, Feb.11	Common Estimators	FS Ch. 1	Data wrangling tutorial lab exercises		HW Set 1 Assign.	
*Feb.16, Feb.18	Working with Probability Dists.	FS Ch. 2	ECDF tutorial lab exercises	No class Feb.16		HW Set 1 Due Feb.18
Feb.23, Feb.25	Working with Prob. Dists. (cont')	FS Ch. 2	Using prob. dists. tutorial lab exercises		HW Set 2 Assign.	
Mar.2, Mar.4	Sampling Distributions	FS Ch. 3, 4.6	Workshop: Simulations and the Bootstrap		HW Set 3 Assign.	HW Set 2 Due Mar.2
Mar.9, Mar.11	Catch-up and Review		Exam I	Exam I		HW Set 3 Due Mar.9
Mar.16, Mar.18	Interval Estimations	FS Ch. 4.1-4.4 GUM: Sect. 2, 3 NS 7.2.6	Workshop: Bootstrap and Tolerance Intervals		HW Set 4 Assign.	
Mar.23, Mar.25	Outlier Detection Methods	NS 1.3.5.17 EPA: Sect 4.4. FDA.	Outlier detection tutorial lab exercises		HW Set 5 Assign.	HW Set 4 Due Mar.23
Mar.30, *Apr.1	Conditional Probability and Testing	FS Ch. 2.1.1-6, pp. 316		No class Apr.1		HW Set 5 Due Mar.30
Apr.13, Apr.15	Conditional Probability and Testing (cont')	FS Ch. 2.1.1-6, pp. 316	Testing tutorial lab exercises		HW Set 6 Assign.	
Apr.20, Apr.22	Hypothesis Testing	FS Ch. 5	Workshop: Bootstrap Hypothesis Testing		HW Set 7 Assign.	HW Set 6 Due Apr.20
Apr.27, Apr.29	Catch-up and Review		Exam II	Exam II		HW Set 7 Due Apr.27
May.4, May.6	Linear Regression	FS Ch. 6	Linear regression tutorial lab exercises		HW Set 8 Assign.	
May.11, May.13	Analysis of Variance	FS Ch. 6	ANOVA tutorial lab exercises		HW Set 9 Assign.	HW Set 8 Due May.11
May.18	PCA, LDA, Bayes Nets			Last day of class		HW Set 9 Due May.18
May XX				Exam III		